

Group No – 46

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Hybrid Retrieval-Augmented Generation (RAG) System

1. Summary

This report documents the execution of a Hybrid Retrieval-Augmented Generation (RAG) system built using a combination of dense retrieval (Sentence Transformers + FAISS), sparse retrieval (BM25), and Reciprocal Rank Fusion (RRF). The system retrieves relevant Wikipedia documents and generates answers using a transformer-based language model (Flan-T5-base).

2. Dataset and Collection

- 200 fixed Wikipedia URLs (stored in fixed_urls.json).
- 300 randomly selected Wikipedia pages (minimum 200 words each).
- Total documents extracted: 490.
- Total chunks created after preprocessing: 1,442.

3. Preprocessing Steps

- Text cleaning (removal of extra whitespace and formatting artifacts).
- Token-based chunking: 400 tokens per chunk.
- Overlap between chunks: 50 tokens.
- Metadata stored: title, URL, chunk ID, chunk index.

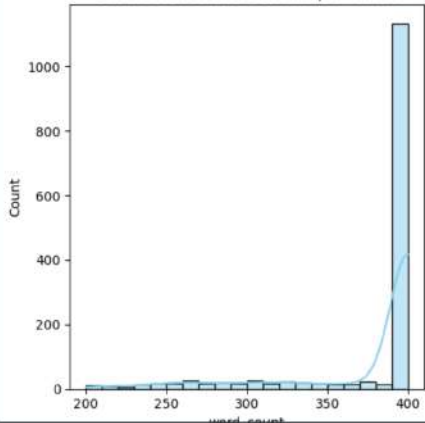
4. Exploratory Data Analysis (EDA)

- Average chunks per URL: approximately 2.94.
- PCA visualization performed on dense embeddings.
- Distribution analysis of word counts and nearest neighbor distances.
- No duplicate chunk IDs detected.

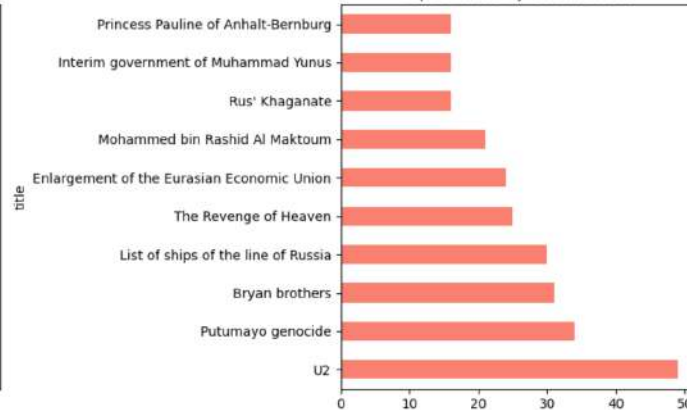
```
Data columns (total 5 columns):
# Column Non-Null Count Dtype
0 chunk_id 1442 non-null object
1 title 1442 non-null object
2 url 1442 non-null object
3 text 1442 non-null object
4 chunk_index 1442 non-null int64
dtypes: int64(1), object(4)
memory usage: 56.5+ KB
None
```

```
### Missing Values ###
chunk_id 0
title 0
url 0
text 0
chunk_index 0
dtype: int64
```

Distribution of Word Counts per Chunk



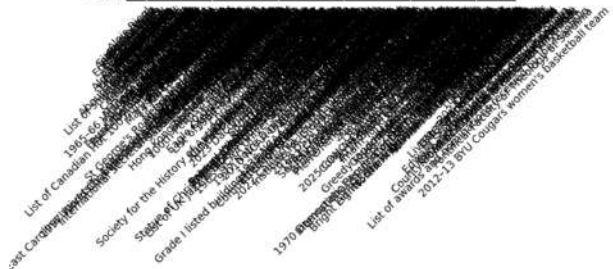
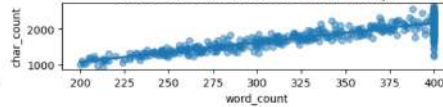
Top 10 Titles by Chunk Count



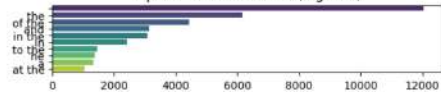
Word Count Variation per Document



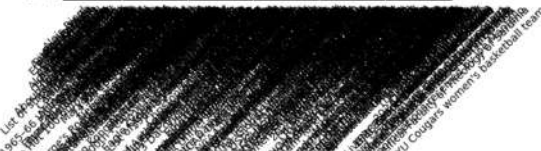
Character Count vs. Word Count (Density)



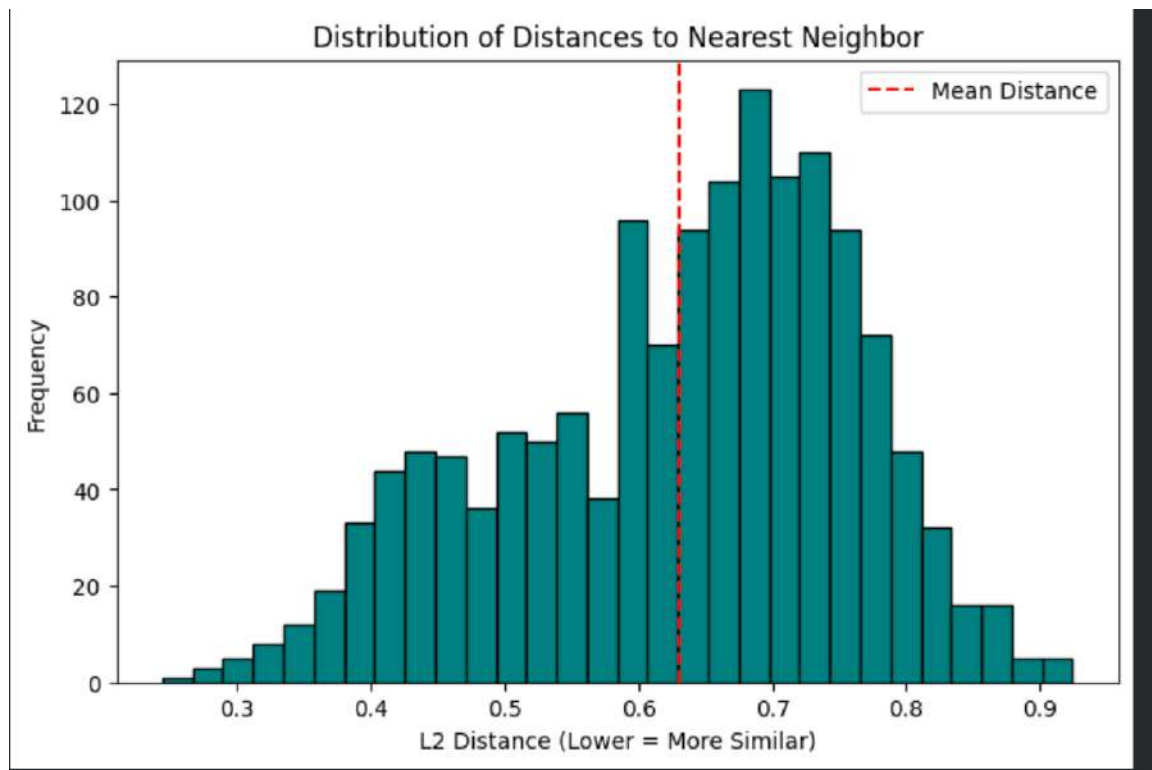
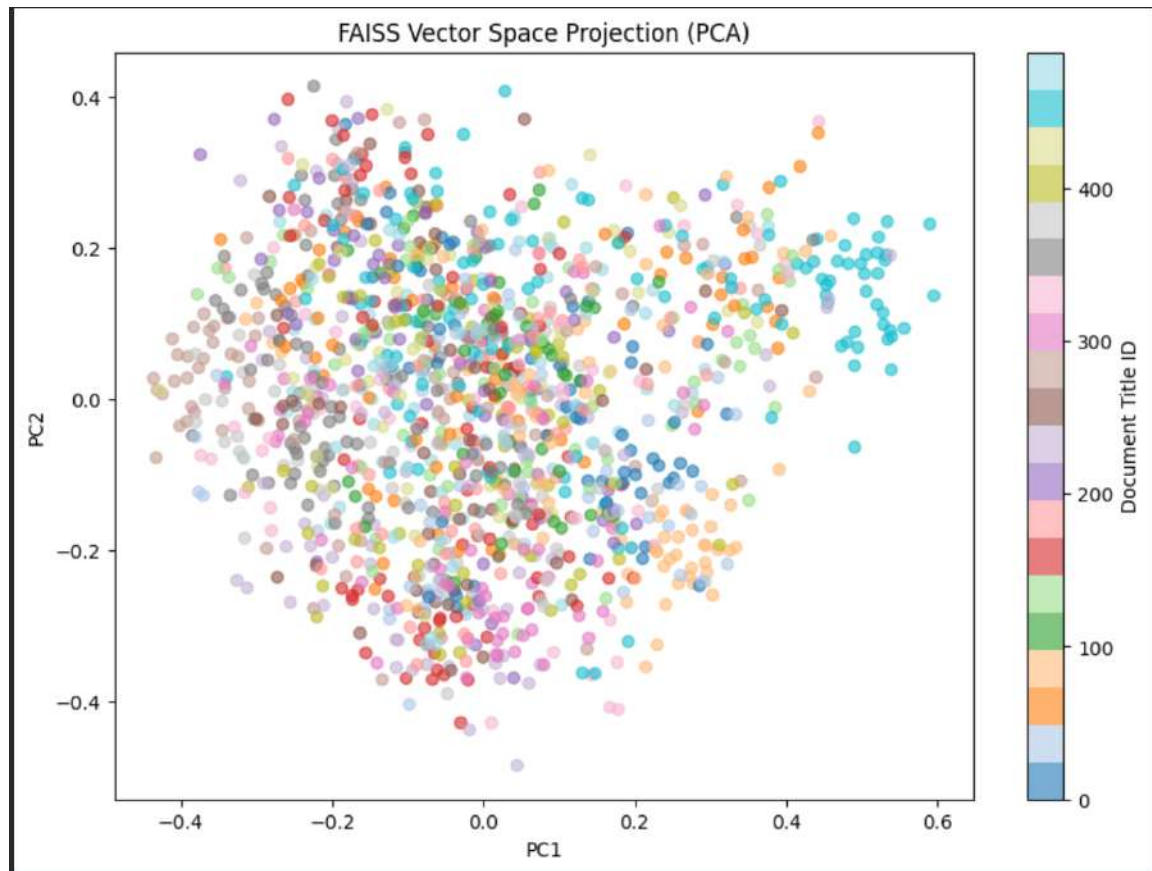
Top 10 Common Phrases (Bigrams)



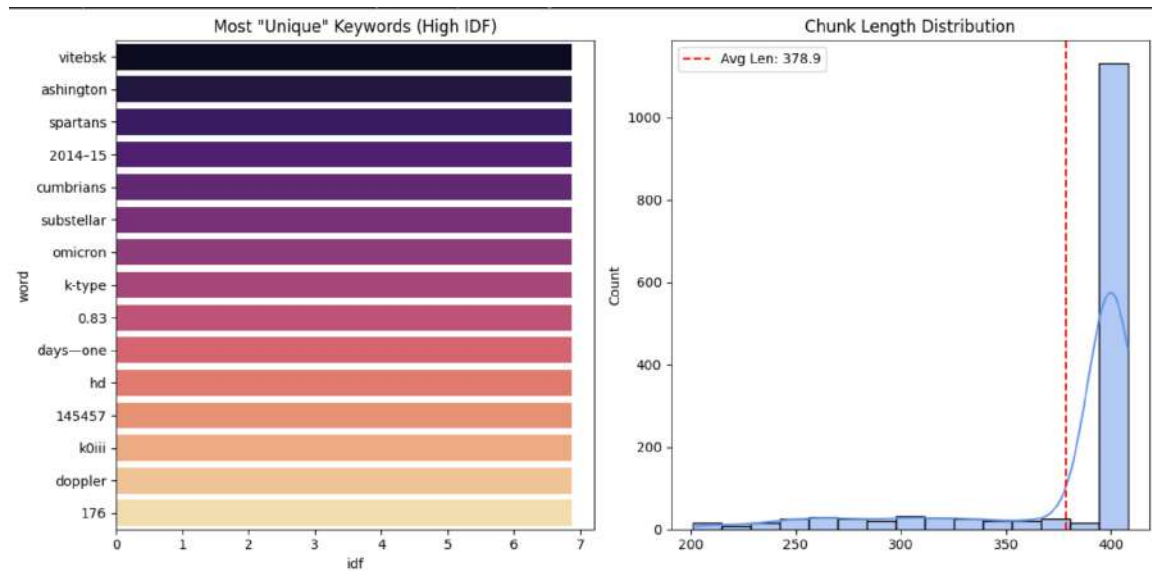
Chunk Index Coverage per Document



Exploratory Data Analysis (EDA) for Dense Vector



Exploratory Data Analysis (EDA) for BM25



5. Retrieval Pipeline

Dense Retrieval:

- Embedding model: all-MiniLM-L6-v2 (Sentence Transformers).
- FAISS IndexFlatIP used for similarity search.
- Embeddings normalized before indexing.

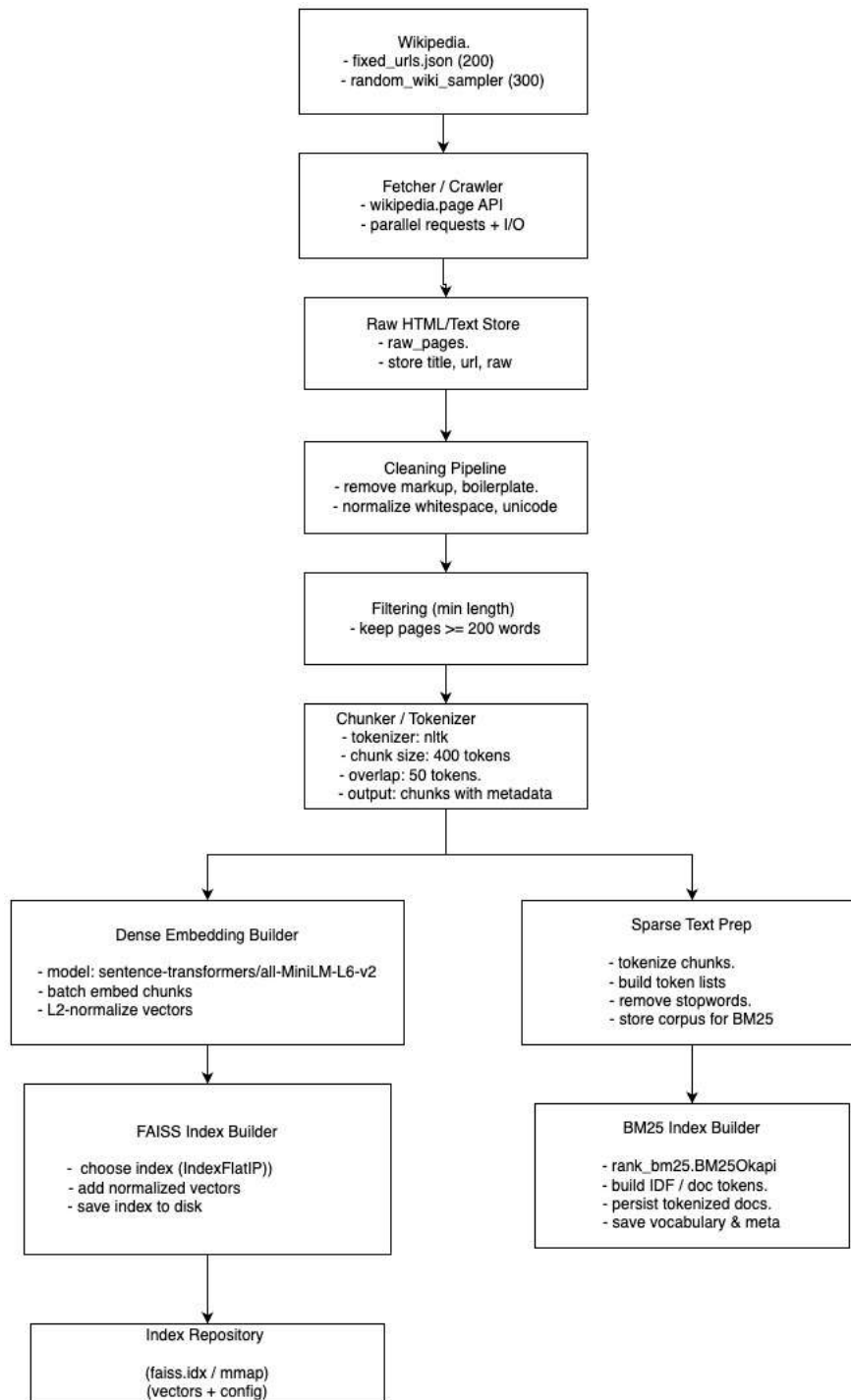
Sparse Retrieval:

- BM25 implemented using rank_bm25 library.
- IDF and document-length distributions analyzed.

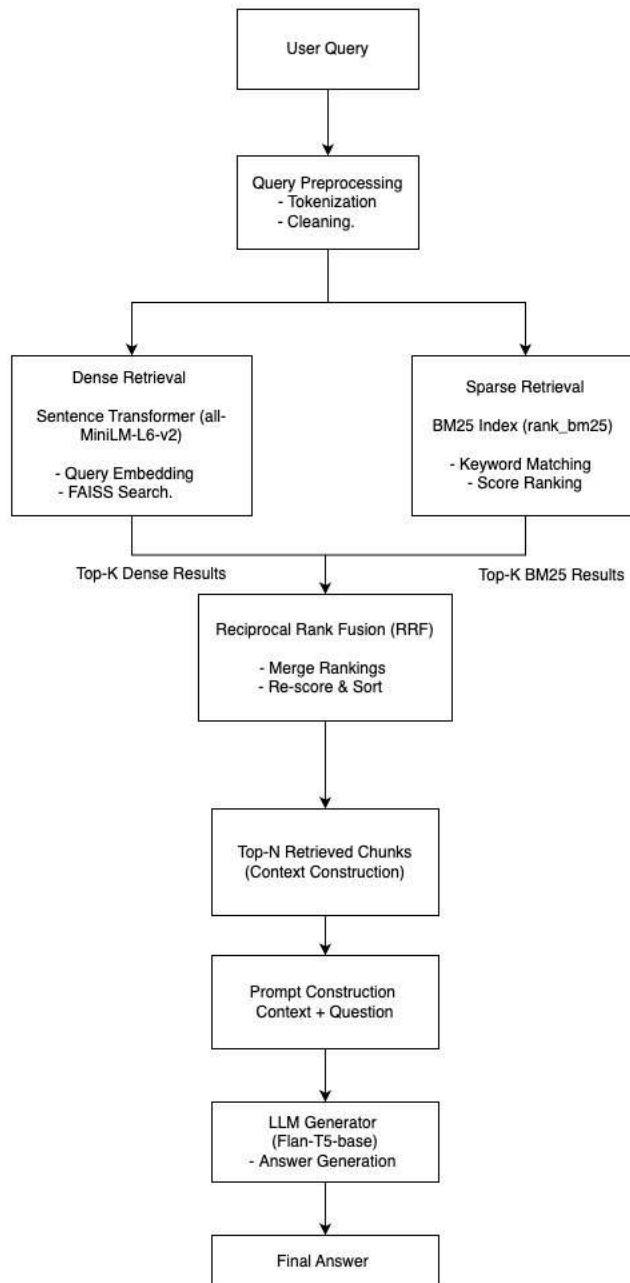
Fusion Method:

- Reciprocal Rank Fusion (RRF) combines dense and sparse rankings.

6. Indexing Pipeline Architecture



7. Retrieval Pipeline Architecture



8. Generation Model

- Model used: google/flan-t5-base.
- Retrieved top-ranked chunks concatenated as context.
- Input truncated to maximum 1024 tokens.
- Final answer generated using sequence-to-sequence decoding.

9. Example Execution

- Query example: 'Tell me about Jim Keogh?'
- Retrieval results compared across Dense, BM25, and RRF.
- Example generation performed for entity-based queries.
- Response time recorded for hybrid retrieval.

User Query

Tell me about Jim Keogh

Clear

Submit

Generated Answer

Jim Keogh is an American technology writer

Top Retrieved Chunks (RRF + Sources)

Chunk 4

Title : James Kavanagh (bishop)

URL : [https://en.wikipedia.org/wiki/James_Kavanagh_\(bishop\)](https://en.wikipedia.org/wiki/James_Kavanagh_(bishop))

RRF Score: 0.0159

Bishop James Kavanagh BA STL Dip Ecom Sci ,MA (Hons) (1914-2002) , was an Irish priest and professor , who served as Auxiliary Bishop in the Dublin Catholic Archdiocese from 1973 until 1991 . == Early life == Kavanagh was born in Dublin in 1914 where he went St. Laurence 's Primary School followed by O'Connell School . He went to Clonliffe Col...

Chunk 5

Title : Mom & Me & Mom

URL : https://en.wikipedia.org/wiki/Mom_%26_Me_%26_Mom

RRF Score: 0.0159

and two cookbooks , Hallelujah ! The Welcome Table in 2004 and Great Food , All Day Long in 2010 . During this period , she was awarded the Lincoln Medal in 2008 and the Presidential Medal of Freedom in 2011 . Angelou had become recognized and highly respected as a spokesperson for Blacks and women and war , as scholar Joseph Braxton has stated , "

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Hybrid RAG System (Dense + BM25 + RRF)

This system retrieves relevant Wikipedia chunks using dense vector search (FAISS), sparse keyword search (BM25), and combines them using Reciprocal Rank Fusion (RRF). The exact prompt sent to the LLM is displayed for transparency.

User Query

who is Manamboor Rajan Babu and what is he the editor of?

Clear

Submit

Generated Answer

Innu the oldest inland-letter literary magazine publishing from Kerala state

Top Retrieved Chunks (RRF + Sources)

Chunk 1

Title : Manamboor Rajan Babu

URL : https://en.wikipedia.org/wiki/Manamboor_Rajan_Babu

RRF Score: 0.0328

Manamboor Rajan Babu (born 1948 October 10) is a Malayalam language poet from Kerala , India . He is the founder and editor of Innu the oldest inland-letter literary magazine publishing from Kerala state . He is the author of fourteen books , including eleven poetry collections . His poems have been translated into English , Hindi , Tamil and oth...

Chunk 2

Title : P. P. Narayanan

URL : https://en.wikipedia.org/wiki/P._P._Narayanan

RRF Score: 0.0161

Former Deputy , Director-General , of the ILO (1989) . On 16 March 1992 , the President of Venezuela , conferred the Order of Francisco de Miranda , First Class , to Dr. P. P. Narayanan for his outstanding services and contributions . The award was received in absentia by a representative in Caracas . == Spiritual side == Narayanan was influenced...

Final Prompt Sent to LLM

[truncated]

10. Observations and Limitations

- Hybrid RRF improves retrieval robustness.
- Context length limited by model token constraints.

Part 2 — Automated Evaluation

Summary

This report documents the execution and results for *Part 2: Automated Evaluation* of the Hybrid RAG system. It summarizes how synthetic QA pairs were generated, the evaluation metrics used (MRR, Hit Rate, Semantic Similarity), and provides a ready-to-run evaluation script, result aggregation steps, visualization suggestions, and recommended next steps.

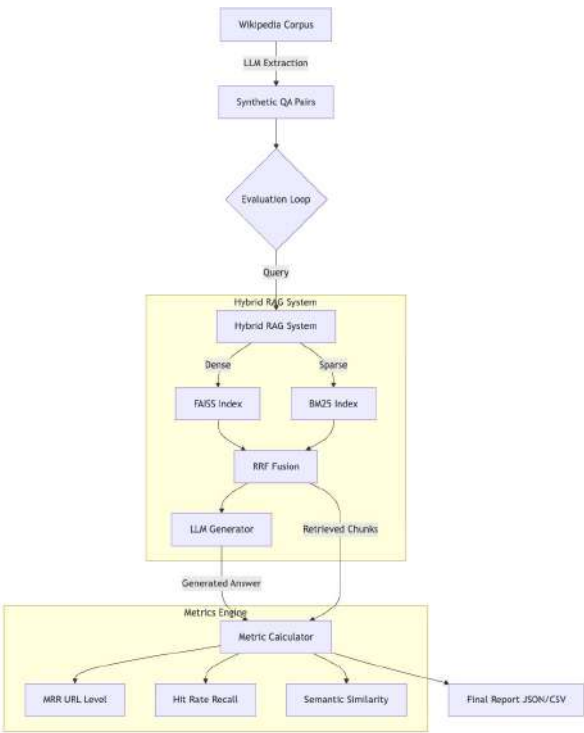
Objectives (Part 2)

- Generate a test set of synthetic questions from the corpus (100 Q&A pairs).
- Evaluate the retrieval and generation pipeline using automatic metrics:
 - Mean Reciprocal Rank (MRR) at URL level
 - Hit Rate (Recall@K)
 - Answer semantic similarity using embeddings
- Provide an automated evaluation loop and visualizations to measure end-to-end quality.

Implementation details

1. **Synthetic QA generation**
 - Randomly sample chunks from chunks (the corpus) and prompt the LLM to generate questions referring to those chunks.
 - Keep the chunk text as the ground-truth answer/context for evaluation.
2. **Evaluation metrics**
 - **MRR (URL-level)**: For each question, find the first retrieved chunk whose url equals the ground-truth URL and take $1 / \text{rank}$. If not found, score = 0.
 - **Hit Rate (Recall@K)**: Binary 1/0 whether the ground-truth URL appears within the retrieved set.
 - **Semantic Similarity**: Encode both the generated answer and the ground-truth chunk text with all-MiniLM-L6-v2 and compute cosine similarity.
3. **Evaluation flow**
 - For each QA pair in qa_dataset (100 items):
 1. Call `hybrid_rag(question, top_k=10, top_n=5)` to get retrieved chunks and a generated answer.
 2. Compute MRR and Hit Rate from the `rrf_chunks` returned.
 3. Compute semantic similarity between the generated answer and the ground-truth chunk text.
 4. Store the per-question metrics and metadata.
 - Aggregate across all queries to compute mean MRR, overall Hit Rate, mean semantic similarity, and produce distributions/boxplots.

Execution Pipeline Architecture



Execution Pipeline Result

